

Conditioning assumptions, MPC key judgements, and indicative projections August 2019

This document contains five tables of projections:

- Table 1 shows the conditioning assumptions used for the MPC's projections;
- Table 2 shows the MPC's mean projections for GDP, CPI inflation and LFS unemployment;
- Table 3 shows a summary of the headline forecast variables;
- Table 4 shows profiles that illustrate the MPC's key judgements as described in Section 5; and
- Table 5 shows indicative projections produced for the MPC by Bank staff to be consistent with the MPC's modal projections for GDP, inflation and unemployment.

The August 2019 projections in Tables 2, 3, 4 and 5 have been conditioned on the paths shown in Table 1, as well as a broad range of asset prices, which embody market expectations of the future stocks of purchased gilts and corporate bonds, and the Term Funding Scheme (TFS) which is financed by the issuance of central bank reserves.

The '*Inflation Report* fan charts August 2019' document and the quarterly market interest rate paths are available from the 'Download the chart slides and data' link at www.bankofengland.co.uk/inflation-report/2019/august-2019.

Table 1 Forecast conditioning assumptions^{(a)(b)}

	Average 1998– 2007	Projections		
		2019	2020	2021
Bank Rate ^(c)	5.0	0.6	0.5	0.5
Sterling effective exchange rate ^(d)	100	76	76	76
Oil prices ^(e)	39	64	63	63
Gas prices ^(f)	29	48	57	57
Nominal government expenditure ^(g)	7¼	5½	3	3¼

Sources: Bank of England, Bloomberg Finance L.P., Eikon from Refinitiv, Office for Budget Responsibility (OBR), ONS and Bank calculations.

(a) The table shows the projections for financial market prices, wholesale energy prices and government spending projections that are used as conditioning assumptions for the MPC's projections for CPI inflation, GDP growth and the unemployment rate.

(b) Financial market data are based on averages in the 15 working days to 24 July 2019. Figures show the average level in Q4 of each year, unless otherwise stated.

(c) Per cent. The path for Bank Rate implied by forward market interest rates. The curves are based on overnight index swap rates.

(d) Index, January 2005 = 100. The convention is that the sterling exchange rate follows a path that is half way between the starting level of the sterling ERI and a path implied by interest rate differentials.

(e) Dollars per barrel. Projection based on monthly Brent futures prices for two quarters, then held flat.

(f) Pence per therm. Projection based on monthly natural gas futures prices for two quarters, then held flat.

(g) Annual average growth rate. Nominal general government consumption and investment. Projections are taken from the OBR's *Economic and Fiscal Outlook* associated with the Spring Statement 2019. Historical data based on NMRP+D7QK.

Table 2 The MPC's mean projections for GDP, CPI inflation and LFS unemployment

	Average 1998– 2007	Projections		
		2019	2020	2021
GDP ^(a)	2.9	1.3	1.3	2.3
CPI inflation ^(b)	1.6	1.6	2.1	2.2
LFS unemployment rate ^(c)	5.3	3.8	3.9	3.6

(a) Annual average growth in real GDP consistent with the mean projection for four-quarter growth in real GDP. The growth rates reported in this table exclude the backcast for GDP.

(b) Four-quarter inflation rate in Q4.

(c) LFS unemployment rate in Q4.

Table 3 Forecast summary^(a)

	Projections			
	2019 Q3	2020 Q3	2021 Q3	2022 Q3
GDP ^(b)	1.0 (1.2)	1.4 (1.7)	2.4 (2.1)	2.5
<i>Including the backcast^(c)</i>	<i>1.0 (1.3)</i>	<i>1.4 (1.7)</i>	<i>2.4 (2.1)</i>	<i>2.5</i>
CPI inflation ^(d)	1.7 (1.8)	1.9 (1.7)	2.2 (2.1)	2.4
LFS unemployment rate	3.7 (3.7)	4.0 (3.9)	3.7 (3.6)	3.3
Excess supply/Excess demand ^(e)	-¼ (-¼)	-¼ (+¼)	+¾ (+¾)	1¾
Bank Rate ^(f)	0.7 (0.7)	0.5 (0.8)	0.5 (0.9)	0.6

(a) Modal projections for GDP, CPI inflation, LFS unemployment and excess supply/excess demand. Figures in parentheses show the corresponding projections in the May 2019 *Inflation Report*.

Projections were only available to 2022 Q2 in May.

(b) Four-quarter growth in real GDP. Excludes the backcast for GDP.

(c) Four-quarter growth in real GDP.

(d) Four-quarter inflation rate.

(e) Per cent of potential GDP. A negative figure implies output is below potential and a positive figure that it is above.

(f) Per cent. The path for Bank Rate implied by forward market interest rates. The curves are based on overnight index swap rates.

Table 4 MPC key judgements^{(a)(b)}

Key Judgement 1: while global activity has weakened and sentiment has deteriorated, looser financial conditions support the return of world growth to its potential rate in the medium term

	Average 1998– 2007	Projections		
		2019	2020	2021
World GDP (UK-weighted) ^(c)	3	2 (2)	2 (2¼)	2¼ (2¼)
World GDP (PPP-weighted) ^(d)	4	3 (3¼)	3¼ (3½)	3½ (3½)
Euro-area GDP ^(e)	2¼	1¼ (1¼)	1½ (1½)	1½ (1¾)
US GDP ^(f)	3	2½ (2¼)	1¾ (1¾)	1¾ (1¾)
Net trade contribution to UK GDP growth ^(g)	-¼	-¾ (-¼)	1¼ (½)	0 (0)

Key Judgement 2: on the conditioning assumption that there is a smooth Brexit, UK demand growth recovers after softening in the near term

	Average 1998– 2007	Projections		
		2019	2020	2021
Business investment contribution to GDP growth ^(h)	¼	-¼ (-¼)	-¼ (¼)	½ (½)
Business investment to GDP ratio ⁽ⁱ⁾	9¾	9 (9)	8¾ (9)	9 (9¼)
Household consumption contribution to GDP growth ^(j)	2¼	1 (1)	1 (1)	1¼ (1¼)
Credit spreads ^(k)	¾ (l)	1¾ (1½)	1¾ (1½)	1¾ (1¾)
Household saving ratio ^(m)	8½	4½ (4½)	3½ (4)	3¾ (4¼)

Key Judgement 3: as GDP growth recovers to above the subdued rate of potential supply growth, excess demand and domestic inflationary pressures build

	Average 1998– 2007	Projections		
		2019	2020	2021
Productivity ⁽ⁿ⁾	2¼	-¼ (0)	1 (1)	1 (1)
Participation rate ^(o)	63	64 (64)	64 (64)	64 (64)
Average hours ^(p)	32¼	32 (32)	32¼ (32¼)	32¼ (32¼)
UK import prices ^(q)	¼	¾ (-½)	½ (0)	¼ (0)
Dollar oil prices ^(r)	39	64 (70)	63 (66)	63 (64)
Unit labour costs ^(s)	2¾	2¾ (2¾)	1¾ (2)	2½ (2½)
Unit wage costs ^(t)	2½	2½ (2¾)	1¾ (2)	2½ (2½)
Private sector regular pay based unit wage costs ^(u)	1¾	3¼ (3)	1¾ (2¼)	2¾ (2½)

Sources: Bank of England, BDRC Continental *SME Finance Monitor*, Bloomberg Finance L.P., British Household Panel Survey, Department for Business, Energy and Industrial Strategy, Eurostat, ICE/BoAML Global Research (used with permission), IMF *World Economic Outlook* (WEO), ONS, US Bureau of Economic Analysis and Bank calculations.

(a) The MPC's projections for GDP growth, CPI inflation and unemployment (as presented in the fan charts) are underpinned by three key judgements. The mapping from the key judgements to individual variables is not precise, but the profiles in the table should be viewed as broadly consistent with the MPC's key judgements.

(b) Figures show annual average growth rates unless otherwise stated. Figures in parentheses show the corresponding projections in the May 2019 *Inflation Report*. Calculations for back data based on ONS data are shown using ONS series identifiers.

(c) Chained-volume measure. Constructed using real GDP growth rates of 180 countries weighted according to their shares in UK exports.

(d) Chained-volume measure. Constructed using real GDP growth rates of 181 countries weighted according to their shares in world GDP using the IMF's purchasing power parity (PPP) weights.

(e) Chained-volume measure. Forecast was finalised before the release of the preliminary flash estimate of euro-area GDP for Q2, so that has not been incorporated.

(f) Chained-volume measure. Forecast was finalised before the release of the advance estimate of US GDP for Q2, so that has not been incorporated.

(g) Chained-volume measure. Exports less imports. Since 1998, exports based on (IKBK-OFNN/(BOKH/BQKO)). Since 1998, imports based on (IKBL-OFNN/(BOKH/BQKO)). Prior to 1998 the data exclude the impact of missing trader intra-community (MTIC) fraud, with net trade based on (IKBK-IBKL). GDP data based on the mode of the MPC's GDP backcast.

(h) Chained-volume measure. Business investment data based on GAN8. GDP data based on the mode of the MPC's GDP backcast. Investment data take account of the transfer of nuclear reactors from the public corporation sector to central government in 2005 Q2.

(i) Chained-volume measure. Annual average. Business investment as a percentage of GDP. GDP data based on ABMI.

(j) Chained-volume measure. Includes non-profit institutions serving households. Household consumption data based on ABJR+HAYO. GDP data based on the mode of the MPC's GDP backcast.

(k) Level in Q4. Percentage point spread over reference rates. Based on a weighted average of household and corporate loan and deposit spreads over appropriate risk-free rates. Indexed to equal zero in 2007 Q3.

(l) Based on the weighted average of spreads for households and large companies over 2003 and 2004 relative to the level in 2007 Q3. Data used to construct the SME spread are not available for that period. The period is chosen as broadly representative of one where spreads were neither unusually tight nor unusually loose.

(m) Percentage of total available household resources. Based on NRJS.

(n) GDP per hour worked. GDP data based on the mode of the MPC's GDP backcast. Hours worked based on YBUS.

(o) Level in Q4. Percentage of the 16+ population. Based on MGWG.

(p) Level in Q4. Average weekly hours worked, in main job and second job. Based on YBUS/MGRZ.

(q) Four-quarter inflation rate in Q4 excluding fuel and the impact of MTIC fraud.

(r) Average level in Q4. Dollars per barrel. Projection based on monthly Brent futures prices for the first two quarters of the forecast period, then held flat.

(s) Four-quarter growth in unit labour costs in Q4. Whole-economy total labour costs divided by GDP at constant prices, based on the mode of the MPC's GDP backcast. Total labour costs comprise compensation of employees and the labour share multiplied by mixed income.

(t) Four-quarter growth in whole-economy unit wage costs in Q4. Whole-economy wage costs divided by GDP at constant prices, based on the mode of the MPC's GDP backcast. Total wage costs are wages and salaries excluding non-wage costs and the labour share multiplied by mixed income.

(u) Four-quarter growth in private sector regular pay based unit wage costs in Q4. Private sector wage costs divided by private sector output at constant prices, based on the mode of the MPC's GDP backcast. Private sector wage costs are average weekly earnings (excluding bonuses) multiplied by private sector employment.

Table 5 Indicative projections consistent with MPC's modal projections^(a)

	Average 1998– 2007	Projections			<i>Memo: mean forecast error^(b)</i>	<i>Memo: level in 2018^(c)</i>
		2019	2020	2021		
Annual average growth rate						
Household consumption ^(d)	3½	1½ (1½)	1½ (1½)	1¾ (1¾)	1¼	1,341
Business investment ^(e)	2½	-2 (-2½)	-1½ (3)	5¼ (5½)	6¼	189
Housing investment ^(f)	3¼	-¾ (-2)	½ (-¾)	3½ (2¾)	7¼	100
Exports ^(g)	4½	1¼ (3¼)	1¾ (1¼)	1¼ (1¼)	3¾	588
Imports ^(h)	6	4¼ (4¼)	-2¼ (0)	1½ (1¼)	3¾	611
Real post-tax labour income ⁽ⁱ⁾	3¼	1¼ (1¼)	½ (1)	2 (2)	2	944
Four-quarter growth rate in Q4						
Employment ^(j)	1	¾ (¾)	½ (½)	1 (¾)	¾	33 ^(k)
Average weekly earnings ^(l)	4¼	3 (3)	3¼ (3½)	4 (3¾)	1	527 ^(m)

(a) These projections are produced by Bank staff for the MPC to be consistent with the MPC's modal projections for GDP growth, CPI inflation and unemployment. Figures show annual average growth rates unless otherwise stated. Figures in parentheses show the corresponding projections in the May 2019 *Inflation Report*. Calculations for back data are shown using ONS series identifiers.

(b) Root mean squared errors of one year ahead forecasts made between February 2003 and February 2018.

(c) £ billion chained-volume measure (reference year 2016) in 2018 unless otherwise stated.

(d) Chained-volume measure. Includes non-profit institutions serving households. Based on ABJR+HAYO.

(e) Chained-volume measure. Based on GAN8.

(f) Chained-volume measure. Whole-economy measure. Includes new dwellings, improvements and spending on services associated with the sale and purchase of property. Based on DFEG+L635+L637.

(g) Chained-volume measure. The historical data exclude the impact of missing trader intra-community (MTIC) fraud. Since 1998 based on IKBK-OFNN/(BOKH/BQKO). Prior to 1998 based on IKBK.

(h) Chained-volume measure. The historical data exclude the impact of MTIC fraud. Since 1998 based on IKBL-OFNN/(BOKH/BQKO). Prior to 1998 based on IKBL.

(i) Wages and salaries plus mixed income and general government benefits less income taxes and employees' National Insurance contributions, deflated by the consumer expenditure deflator. Based on [(ROY)+ROYH-(RPHS+AIIV-CUCT)+GZVX]/[(ABJQ+HAYE)/(ABJR+HAYO)].

(j) Based on MGRZ.

(k) People aged 16+ employed in 2018 Q4, millions.

(l) Whole-economy total pay. Growth rates since 2001 based on KAB9. Prior to 2001, growth rates are based on historical estimates of AWE, with ONS series identifier MD9M.

(m) Average pounds sterling earned per week in 2018 Q4.